

DELETE Data

Learning Objectives

- Write DELETE statements to remove specific records from a table
- Use WHERE clauses to prevent accidental deletion of all data
- Explain the difference between DELETE and DROP commands

Engage (5-7 minutes)

When a student graduates, their record is no longer needed in the current class register. Removing it keeps the database clean and efficient. The DELETE command removes records permanently from a table.

Explore (10-12 minutes)

Given a Students table, write a DELETE statement to remove the student with ID 10. Then write a DELETE statement without WHERE and observe the warning. Compare DELETE (removes rows) with DROP TABLE (removes the entire table structure).

Explain (10-12 minutes)

DELETE removes specific rows from a table when filtered by a WHERE clause. Without WHERE, all rows are deleted but the table structure remains. DROP TABLE removes the table entirely including its structure. TRUNCATE is a faster alternative to DELETE for removing all rows while preserving the table structure.

Elaborate (8-10 minutes)

A company removes an inactive employee. Write a DELETE statement for that employee. Discuss backup strategies before deletion and explain how foreign key constraints might prevent deleting a record that is referenced by other tables.

Evaluate (5-8 minutes)

Write a DELETE statement to remove all students with Grade “F” from a Students table. Explain three differences between DELETE, TRUNCATE, and DROP commands.